

## Final Week Deliverable Report: Healthcare Persistency Project

### Team:

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### Problem Understanding:

This project focuses on understanding factors influencing patient drug persistency, specifically within ABC Pharma Company. Persistency is critical in improving patient outcomes, particularly in chronic treatments where adherence impacts overall effectiveness. The goal was to develop predictive models to identify at-risk patients, enabling targeted interventions to enhance drug adherence and patient support.

### Approach and Solution:

The goal of this project is to explore multiple predictive models, including Logistic Regression, Random Forest, and XGBoost, to assess their performance in identifying persistent vs. non-persistent patients based on a dataset comprising demographic details, comorbidities, and treatment factors. Each model was evaluated for accuracy, recall, and precision to ensure reliable insights for ABC Pharma's needs.

1. **Logistic Regression** achieved an accuracy of 80.8%, showing balanced performance in identifying persistent and non-persistent cases with an emphasis on interpretability.
2. **Random Forest** reached an accuracy of 79.8%, offering valuable insights into feature importance but slightly lower recall on persistent patients.
3. **XGBoost** demonstrated an 80.2% accuracy, excelling in recall for persistent patients, making it suitable for applications prioritizing sensitivity.

### Final Recommendation:

Based on the EDA, it suggests Logistic Regression for its interpretability and balanced performance, making it suitable for practical deployment where explainability is crucial. However, XGBoost could also be considered in contexts where high sensitivity to persistent patients is prioritized.

**GitHub Repository Link:**

<https://github.com/shuyi95/HealthcareDrugPersistancy>

By implementing these predictive models, ABC Pharma can proactively address patient adherence issues, ultimately leading to improved healthcare outcomes and optimized treatment plans.